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Colorimetric and Fluorimetric Assays to Quantitate Micromolar Concentrations of Transition Metals[1]

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Transition metal ions, although maintained at low concentrations, play diverse important roles in many biological processes. Two assays useful for the rapid quantification of a range of first-row transition metal ions have been developed. The colorimetric assay extends the 4-(2-pyridylazo)resorcinol assay of Hunt *et al.* (*J. Biol. Chem.* 255, 14793 (1984)) to measure nanomole quantities of Co^{2+}, Ni^{2+}, and Cu^{2+} as well as Zn^{2+}. The fluorimetric assay takes advantage of the coordination of a number of metal ions (Mn^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Zn^{2+}, Cd^{2+}) by Fura-2 and can also be used to measure nanomole quantities of these ions. The assays developed here have the advantage of not requiring the extensive sample preparation necessary for other methodologies, such as atomic absorption spectroscopy and inductively coupled plasma emission spectroscopy (ICPES), while being comparable in accuracy to the detection limits of ICPES for the first-row transition metal ions. To demonstrate the effectiveness of these assays, we determined the affinity of carbonic anhydrase II (CA II), a prototypical zinc enzyme, for Ni^{2+} and Cd^{2+}. These data indicate that CA II binds transition metals with high affinity and is much more selective for Zn^{2+} over Ni^{2+} or Cd^{2+} than most small-molecule chelators or other metalloenzymes. © 2000 Academic Press

Key Words: **4-(2-pyridylazo)resorcinol; PAR; Fura-2; transition metal; assay; carbonic anhydrase.**

The first-row transition metal ions (Mn^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, and Zn^{2+}) play a number of essential biological roles (1). In particular, these metal ions are cofactors in a majority of the cellular metalloproteins. The concentrations of transition metal ions and the factors governing the insertion of the correct metal ion into metalloproteins have implications for a large number of disciplines, including limnology and oceanography (2), enzymology (3), protein design (4), toxicology (5), and human genetic diseases (e.g., Menkes and Wilson's diseases) (6).

Development of a rapid assay for the determination of nanomoles of several different metal ions would enhance the investigation of many biological problems, including features governing metal ion selectivity in proteins. Currently, the methods to rapidly measure nanomoles of first-row transition metals are limited and often require extensive sample preparation (7–9). Previous studies measuring the affinity of a protein for a range of metal ions relied on different assays for each metal ion, including radioactive assays (10), atomic absorption spectroscopy (8, 11), or competition experiments with an easily measured metal ion (12–14). Each of these techniques has disadvantages, such as the use of high-energy radioactive material or significant sample preparation prior to the assay. Additionally, the use of multiple assays or multiple ions in competition can potentially generate systematic errors that can reduce the comparative accuracy of the data.

Therefore, an ideal assay to determine metal ion specificity in proteins would be rapid, require little or no preprocessing of the sample, and be able to detect small amounts of a range of metal ions with high precision. Optical sensing methods have the advantage of speed, simplicity, and sensitivity. A number of optical methods have been previously developed to quantitate the concentration of a single metal ion. In particular, Hunt and colleagues (15) developed a widely used colorimetric assay that measures micromolar concentrations (nanomolar quantities) of Zn^{2+} from the

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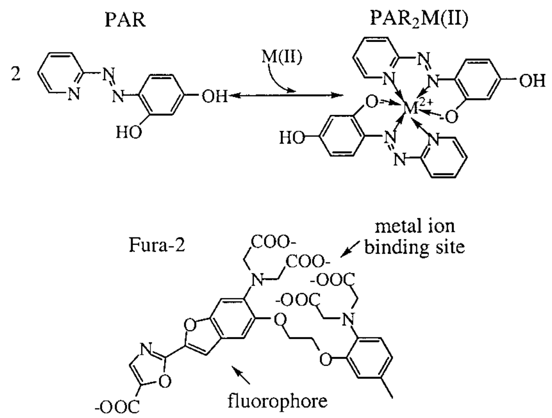

FIG. 1. Structures of PAR and Fura-2. PAR binds transition metal ions in a 2:1 complex through the pyridyl nitrogen, the distal nitrogen of the azo moiety, and the 3-phenolate oxygen of the resorcinol moiety (17). Fura-2 binds metals in a 1:1 complex with the four carboxylate groups. The increase in fluorescence of Fura-2 upon metal coordination is proposed to be due to the relief of quenching of the fluorophore by the electron-rich metal ion binding site moiety (48).

alteration of the absorption spectrum of 4-(2-pyridylazo)resorcinol (PAR,[3] Fig. 1) upon zinc chelation (16–19). However, this method has not been examined for quantitating other transition metals. Alternatively, fluorimetric changes may be used to monitor metal ion concentrations. The fluorophore Fura-2 (Molecular Probes) is widely used as a calcium indicator (20–22) by monitoring fluorescence increases upon the formation of a Ca^{2+}–Fura-2 complex (Fig. 1). Although Fura-2 is known to bind Zn^{2+} more tightly than Ca^{2+} (22), this fluorophore has not previously been used extensively to measure the concentration of transition metals.

Here we develop optical assays to measure the concentration of a number of transition metals using a single method. First, we demonstrate that the colorimetric PAR assay can be used to measure nanomoles of Co^{2+}, Ni^{2+}, Cu^{2+}, and Zn^{2+} but not Mn^{2+} or Cd^{2+}. Second, we show that the fluorescent Fura-2 assay can be extended to accurately detect nanomoles (micromolar concentrations) of Mn^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Zn^{2+}, and Cd^{2+}. This is the first demonstration that PAR may be used to directly quantitate metals other than zinc and the first demonstration that Fura-2 may be used to directly quantitate ions other than calcium. Both of these assays are rapid, requiring little sample preparation, and can be easily used to quantify metal ion concentrations from 1 to 10 μM. Finally, we demonstrate the utility of these assays by measuring the metal ion specificity of the zinc metalloenzyme human carbonic anhydrase II (CA II).

[3] Abbreviations used: AA, graphite furnace atomic absorption spectroscopy; CA II, human carbonic anhydrase II; DPA, dipicolinic acid; ICPES, inductively coupled plasma emission spectroscopy; ICP–HRMS, inductively coupled plasma–high-resolution mass spectrometry; Mops, (*N*-morpholino)propanesulfonic acid; PAR, 4-(2-pyridylazo)resorcinol.

MATERIALS AND METHODS

Preparation of plasticware and solutions. All plasticware was acid-washed. Thereafter the plasticware was soaked in a 4 mM EDTA solution overnight and rinsed thoroughly with deionized water (18 MΩ) before use. All solutions were prepared in plasticware using deionized water (18 MΩ) and metal-free pipette tips (Bio-Rad) to minimize metal ion contamination. All reagents, buffers, and solvents were of the highest purity available. The Mn^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, and Cd^{2+} metal ion solutions used in the standard curves and metal ion buffer solutions were atomic absorption standards (~1000 μg/ml in 1 wt% HNO_3, Aldrich). The Zn^{2+} metal ion solution was a $ZnSO_4$ volumetric standard (0.0499 M solution in water, Aldrich).

Preparation of metal ion buffers. Metal ion buffers used in protein–metal ion affinity experiments were prepared with bicine (Aldrich) or citric acid (Mallinckrodt). Additionally, trace amounts (4–6 μM) of dipicolinic acid (DPA, Sigma) were included to ensure rapid equilibration of the metal ion with the protein (23). The pH of the solution was adjusted to 7.0 with semiconductor-grade NaOH (99.99%, Aldrich), and 3-(*N*-morpholino)propanesulfonic acid (Mops, Research Organics, Inc.) was added to maintain the total combined buffer concentration at 15 mM. The buffer combinations and concentrations were adjusted such that >90% of the total metal ion was chelated by the buffers. The concentration of free metal ion at a particular pH and buffer concentration was calculated from the known stability constants of the buffers (24).

Colorimetric metal ion assay. To determine metal ion concentrations, first either the metal standard solution (for the standard curve) or the protein solution (for the experimental data) is mixed with guanidine–HCl (4 M, final concentration) and incubated at room temperature for 1–3 min (25). A freshly prepared solution of PAR (final concentration 100 μM) is then mixed with the sample and the absorbance is immediately determined at 500 (Ni^{2+} and Zn^{2+}) or 514 nm (Co^{2+} and Cu^{2+}) using a Shimadzu UV-265FW spectrophotometer with quartz cuvettes of 1.00-cm path length. A standard curve (0–10 μM) was assayed with each series of experimental samples. The absorption spectra of PAR and PAR–metal complexes were determined from 250 to 450 nm using the same conditions.

Fluorescent metal ion assay. The fluorescence emission and excitation spectra of Fura-2 (final concentration 20 μM) and Fura-2–metal solutions were measured in 7.5 mM Mops, 4 M guanidine–HCl, pH 7.0, using an Aminco Bowman Series 2 luminescence spectrophotometer with quartz cuvettes and emission and excitation slit widths of 4 nm. For the emission spectra, the excitation wavelength was 350 nm and for the excitation spectra, the emission wavelength was 500 nm.

The fluorescence assay is very similar to the colorimetric assay. The sample or metal standard solution is diluted into a final concentration of 4 M guanidine–HCl, and the solution is vortexed and incubated at room temperature for several minutes before 20 μM (final concentration) of Fura-2 (Molecular Probes) is added. The fluorescence is determined at an emission wavelength of 510 (Mn^{2+}, Co^{2+}, Zn^{2+}, Cu^{2+}), 500 (Ni^{2+}), or 465 nm (Cd^{2+}) and at an excitation wavelength of 340 nm. The metal ion concentration in the samples was determined by comparison to a standard curve (0–10 μM). Also, as in the colorimetric assay, either equimolar apoenzyme was included in the standard curve or several protein concentrations were examined.

Preparation of CA II and apoenzyme. The CA II protein was recombinantly expressed and purified as previously described (26). Fresh apoenzyme ($\sim 5 \times 10^{-7}$ mol) was prepared by Amicon diaflow filtration (27) against several 50-ml washes of 50–100 mM DPA, 15 mM Mops, pH 7.0, followed by several 30-ml washes of 15 mM Mops buffer, pH 7.0, followed by chromatography on a PD-10 column (Sephadex G-25M, 5 cm $\times$ 15 cm, Pharmacia) preequilibrated with 15 mM Mops buffer, pH 7.0. The enzyme concentration was then determined using the CA II extinction coefficient (ϵ_{280} = 54,000 M^{-1} cm^{-1}) (28). The bound zinc ion concentration was determined by the PAR assay (see above) and the apoenzyme preparation was used only if $\geq$90% of the bound metal was removed.

Determination of binding affinity. Dissociation constants (K_D values) for metals were determined by dialyzing apo-CA II (0.5 ml of 60–80 μM protein) against various concentrations of free metal, maintained by metal-chelator buffers, in a background of $\geq$4 μM DPA, at pH 7.0, for 16–22 h at 30°C. Concentrations of free metal ions were calculated as described above. The dialysis bags (Spectra/Por metal-free CE membranes, molecular weight cutoff 15,000) were rinsed thoroughly with deionized water (18 MΩ). The total metal ion concentration was maintained at >40-fold above the total protein concentration. After equilibration (16–22 h), unbound metal ions were removed from the sample solutions by gel filtration over a PD-10 column (Sephadex G-25M, 5 cm $\times$ 15 cm, Pharmacia). The enzyme concentration ($[E]_{tot}$) was determined from absorbance (28) and the bound metal ion concentration ($[E \cdot M]$) was measured by either the PAR colorimetric assay (Ni^{2+}) or the fluorescent Fura-2 assay (Cd^{2+}), as detailed above. The dissociation constant and asymptotic standard error were calculated using the program KaleidaGraph (Synergy Software) to fit the data to Eq. [1], where C varied from 0.9 to 1.1.

$$[E \cdot M]/[E]_{tot} = C/(1 + K_D/[M]_{free}) \qquad [1]$$

RESULTS

There is currently no single, rapid, in-solution assay for the direct determination of nanomoles of most of the first-row transition metal ions. Here, we develop such an assay to rapidly measure nanomoles of Mn^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Zn^{2+}, and Cd^{2+} using either PAR or Fura-2 as a metallochromic indicator. PAR (Fig. 1) forms water-soluble complexes with a large number of metal ions (18) with a 2:1 stoichiometry. The coordination by Zn^{2+} alters the absorbance spectrum of PAR, leading to the use of this indicator in a colorimetric zinc assay (16). Additionally, PAR has previously been used to quantify Ni^{2+} and Co^{3+} in a capillary electrophoresis and UV detection assay (17).

Colorimetric assay. To determine whether a colorimetric assay with PAR could also be used to directly quantitate concentrations of Mn^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, and Cd^{2+}, as well as Zn^{2+}, the absorption spectra of PAR in the presence and absence of the metal ions (Fig. 2) were measured. The measurements were repeated in the presence of apo-CA II, verifying that the presence of protein did not significantly affect the spectrum (data not shown). The spectrum of PAR alone from 250 to 650 nm has a single absorption band at 415 nm. The addition of 10 μM Cd^{2+} (Fig. 2A) or Mn^{2+} (data not shown) does not significantly alter the spectrum of PAR. Therefore, this indicator cannot be used to assay concentrations of these metals. However, the addition of micromolar concentrations of Co^{2+}, Ni^{2+}, Cu^{2+}, or Zn^{2+} causes a decrease in the PAR absorbance at 415 nm concomitantly with an increase in the absorbance at higher wavelengths (Fig. 2A). The shape and maximal absorption peak of the PAR $\cdot$ M complexes is somewhat dependent on the transition metal. The difference absorption spectra (e.g., Fig. 2B, PAR $\cdot$ Ni^{2+} − PAR) clearly reveal that the maximum decrease in absorption occurs at 410 nm for each metal but that the greatest increase in absorbance is metal dependent, occurring at 500 nm for Ni^{2+} and Zn^{2+} or at 514 nm for Co^{2+} and Cu^{2+} (Table 1).

To determine whether these absorbance changes can be a basis for a metal ion assay, we investigated the absorbance as a function of metal ion. At 100 μM PAR, the absorbance changes at 410 and 500 nm are linearly dependent upon the concentration of added Ni^{2+} up to 10 μM (Fig. 2B). Similarly, the absorbance changes are linearly dependent on the concentration of Co^{2+}, Ni^{2+}, Cu^{2+}, and Zn^{2+}, up to 10 μM. The difference extinction

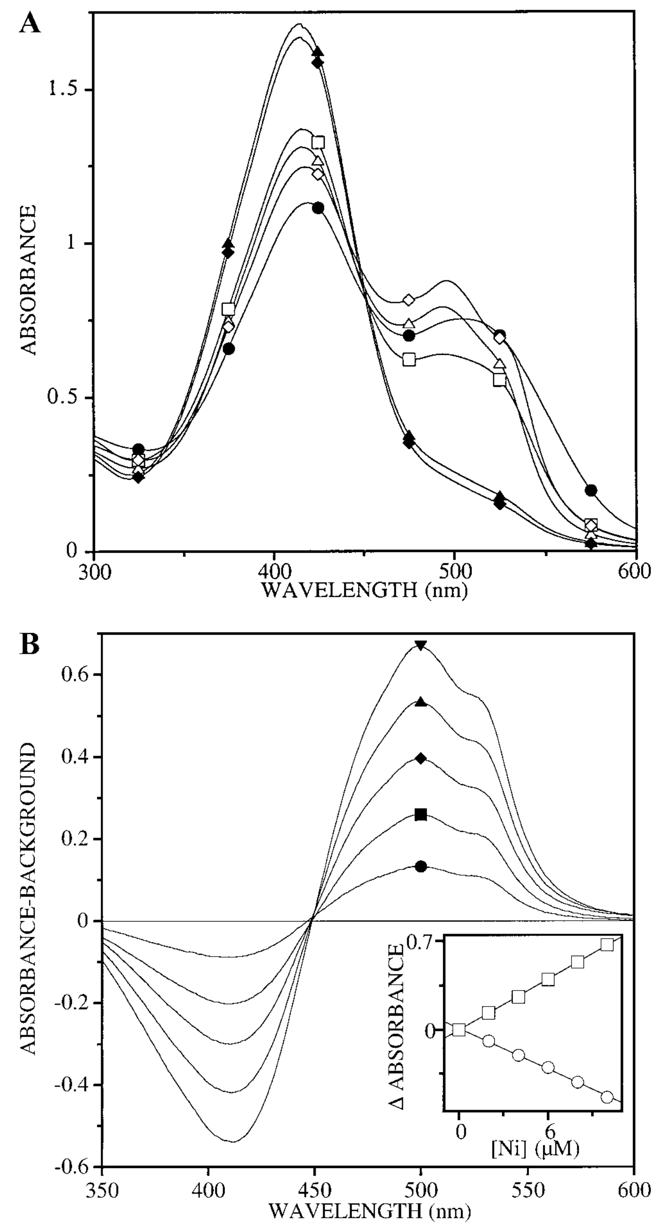

FIG. 2. (A) The absorption spectrum of 100 μM PAR was measured in 4 M guanidine–HCl, 7.5 mM Mops, pH 7.0 at 25°C, in the absence of added metal (◆) and after addition of 10 μM of Co^{2+} (●), Ni^{2+} (◇), Cu^{2+} (□), Zn^{2+} (△), or Cd^{2+} (▲). The spectrum of PAR in the presence of 10 μM Mn^{2+} is not shown since it is nearly identical to the spectrum of PAR with 10 μM Cd^{2+}. (B) The absorption spectrum of 100 μM PAR was measured in 4 M guanidine–HCl, 7.5 mM Mops, pH 7.0 at 25°C, in the absence of added metal and after addition of varying concentrations of Ni^{2+}. Compared to the absorbance of PAR, the Ni^{2+}–PAR complex showed a maximum at 515 nm (ϵ_{515} = ~67,000 M^{-1} cm^{-1}), a minimum at 410 nm (ϵ_{410} = ~−54,000 M^{-1} cm^{-1}), and an isosbestic point at 450 nm. The change in absorbance at 500 (□) or 410 (○) nm is linearly dependent on [Ni^{2+}] up to at least 10 μM (see inset).

coefficient is large and comparable for Co^{2+}, Ni^{2+}, and Zn^{2+} and only decreased slightly for Cu^{2+} (Table 1), allowing for rapid and reproducible determination of micromolar concentrations of these metal ions. The change in extinction coefficient ($\Delta\epsilon$) with increasing concentration of Mn^{2+} or Cd^{2+} is 10-fold lower than for the other ions, preventing the use of PAR to accurately quantitate micromolar concentrations of Mn^{2+} or Cd^{2+}. Therefore, another indicator is required.

Fluorimetric assay. The fluorescent indicator Fura-2 (Fig. 1), used widely to determine calcium concentration, coordinates a wide range of ions at nanomolar concentrations, including Ca^{2+} (K_D = 135 nM), Mn^{2+} (3.2 nM), Fe^{2+} (45 nM), and Zn^{2+} (1.6 nM) (22). To investigate whether a fluorescence change in Fura-2 could be used to directly quantitate micromolar concentrations of transition metal ions, we measured the excitation and emission spectra of Fura-2 and Fura-2–metal ion complexes (Fig. 3), using excess (20 μM) Fura-2.

The spectrum of Fura-2 alone has an excitation peak at 350 nm with a shoulder at 390 nm and an emission peak at 495 nm with a shoulder at 510–525 nm. Consistent with previous data (21, 22, 29–31), the fluorescence of Fura-2 is quenched by Mn^{2+}, Co^{2+}, Ni^{2+}, or Cu^{2+} and enhanced by Zn^{2+} and Cd^{2+}. Furthermore, Mn^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, and Zn^{2+} cause a 5-nm red shift in the excitation wavelength, to 355 nm, and a decrease in the shoulder on the excitation peak. However, in the emission spectra, addition of these metals mainly alters the fluorescence intensity while leaving the peak shape and wavelength unperturbed. In both the excitation and emission spectra, the fluorescence signals are quenched to identical levels by the addition of 10 μM Cu^{2+} or 10 μM Ni^{2+} (e.g., the emission fluorescence is 76% of the original fluorescence at 500 nm) and quenched to a greater extent by 10 μM Mn^{2+} or Co^{2+} (e.g., the emission fluorescence is 57% of the original fluorescence at 500 nm) (Fig. 3, Table 2). These spectra are unaffected by the addition of denatured apo-CA II up to 10 μM total protein concentration (data not shown). Addition of either 10 μM Zn^{2+} or 10 μM Cd^{2+}

TABLE 1

Difference Spectra of PAR–Metal Complexes: Wavelength Maxima and Extinction Coefficient[a]

Metal	$\lambda_{\Delta max}$ (nm)	ΔAbs/μM metal[b]	$\Delta\epsilon$ (M^{-1} cm^{-1})
Mn^{2+}	500	0.002 ± 0.0002	2,000
Co^{2+}	514	0.050 ± 0.0001	50,000
Ni^{2+}	500	0.052 ± 0.0003	52,000
Cu^{2+}	514	0.032 ± 0.0002	32,000
Zn^{2+}	500	0.046 ± 0.0001	46,000
Cd^{2+}	500	0.003 ± 0.0002	3,000

[a] Measured in 100 μM PAR, 4 M guanidine–HCl, 7.5 mM Mops, pH 7.0, 25°C.

[b] The asymptotic standard error was calculated from a least-squares linear fit of the data, using the program KaleidaGraph (Synergy Software).

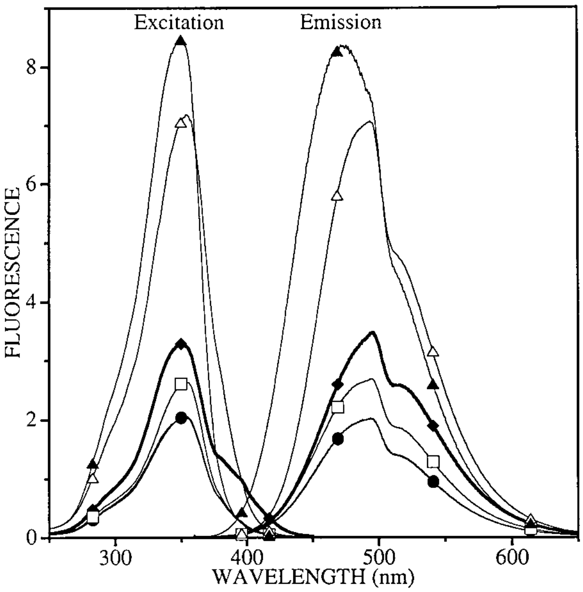

FIG. 3. The fluorescence excitation and emission spectra of 20 μM Fura-2 were measured in 4 M guanidine–HCl, 7.5 mM Mops, pH 7.0 at 25°C, using an excitation wavelength of 340 nm, an emission wavelength of 510 nm, and excitation and emission slit widths of 4 nm, in the absence of added metal (◆) and after addition of 10 μM of Co^{2+} (●), Cu^{2+} (□), Zn^{2+} (△), or Cd^{2+} (▲). The spectra of Fura-2 in the presence of either 10 μM Mn^{2+} or 10 μM Ni^{2+} are not shown since they are nearly identical to the spectrum of Fura-2 with 10 μM Co^{2+} or 10 μM Cu^{2+}, respectively.

enhances both the observed emission and excitation fluorescence of Fura-2 (e.g., 2-fold enhancement of original emission fluorescence at 500 nm for Zn^{2+} and 2.5-fold enhancement of original emission fluorescence at 475 nm for Cd^{2+}) and decreases the prominence of the shoulder on the excitation peak. While the shape and the maximum wavelength of the fluorescence emission spectrum of Fura-2 · Zn are similar to those of Fura-2 alone, addition of 10 μM Cd^{2+} causes a 20-nm blue shift to 475 nm and decreases the prominence of the shoulder at 510–525 nm. Furthermore, unlike the other metal ions, the wavelength of the fluorescence excitation peak is unperturbed by the addition of 10 μM Cd^{2+}. While these changes in fluorescence are useful for measuring metal ion concentrations, the spectral changes are not sufficient to allow identification of the transition metal ion.

To determine whether Fura-2 can be used to accurately measure metal ion concentrations, we measured fluorescence as a function of the metal ion concentration (Fig. 4). The excitation and emission wavelengths were chosen to optimize the change in the fluorescence signal upon addition of the metal ion. Furthermore, the accuracy of the assay was enhanced by using a ratiometric method in which the fluorescence determined at the maximal excitation and emission wavelengths of a specific metal–Fura-2 solution was divided by the fluorescence observed at the isosbestic excitation wavelength (380 nm for Mn^{2+}, Co^{2+}, Ni^{2+}, Zn^{2+}, and Cu^{2+} or 385 nm for Cd^{2+}) of the same metal–Fura-2 solution. The observed fluorescence changes in this ratiometric method are independent of fluctuations in the dye concentration, path length, and instrument sensitivity (22). In each case, the fluorescence emission is linearly dependent on the concentration of added metal. Therefore, Fura-2 can be used to directly quantify all of these metal ions. However, the change in fluorescence upon chelation of Zn^{2+}, Cd^{2+}, Co^{2+}, and Ni^{2+} is typically lower than the change in absorbance seen with the PAR assay and therefore the PAR assay is more sensitive for these metals. For this reason, the colorimetric assay is preferred for the concentration determinations of Co^{2+}, Ni^{2+}, Cu^{2+}, and Zn^{2+} while the fluorimetric assay is preferred for the concentration determinations of Mn^{2+} and Cd^{2+}.

Metal specificity of CA II. To demonstrate the utility of these assays, we determined the binding affinity of human CA II for Ni^{2+} and Cd^{2+} using an equilibrium dialysis method. For CA II, equilibration with zinc can be slow (32) and, therefore, we included low concentrations of DPA, which effectively catalyzes metal ion dissociation from CA II (23) in our standard conditions. Examination of the measured zinc dissociation constant after an equilibration time of 16 h and after an equilibration time of 1 week demonstrated that 16 h was sufficient for full equilibration (data not shown).

TABLE 2

Fluorescence of Fura-2–Metal Complexes: Wavelength Maxima[a]

Metal	$\lambda_{\Delta max}$ (nm) Emission	Excitation	Assay $(Fl/Fl_0)/[M]$ (μM^{-1})
Mn^{2+}	495	350	-0.074 ± 0.001[b]
Co^{2+}	495	350	-0.063 ± 0.001[b]
Ni^{2+}	500	340	-0.097 ± 0.001[b]
Cu^{2+}	500	340	-0.097 ± 0.001[b]
Zn^{2+}	480	360	0.063 ± 0.006[c]
Cd^{2+}	465	350	0.133 ± 0.004[c]

[a] Measured in 20 μM Fura-2, 4 M guanidine–HCl, 7.5 mM Mops, pH 7.0, 25°C.

[b] The wavelengths used for the assays were $\lambda_{Ex} = 340$ nm (Mn^{2+}, Co^{2+}, Ni^{2+}, and Cu^{2+}) and $\lambda_{Em} = 510$ or 500 nm (Ni^{2+} only). These wavelengths do not result in $(Fl/Fl_0)/[M]$ values that differ greatly from those found using the $\lambda_{\Delta max}$ wavelengths but have the advantage of not being on a steep slope in the spectrum, increasing reproducibility slightly. The asymptotic standard error was calculated from a least-squares linear fit of the data, using the program KaleidaGraph (Synergy Software).

[c] The wavelengths used for the assays were $\lambda_{Ex} = 380$ (Zn^{2+}) or 373 nm (Cd^{2+}) and $\lambda_{Em} = 510$ (Zn^{2+}) or 465 nm (Cd^{2+}). The slope ($\Delta Fl/[M]$) is decreased slightly using these wavelengths rather than the wavelengths of maximal ΔFl; however, the accuracy is enhanced due to the decrease in light scattering.

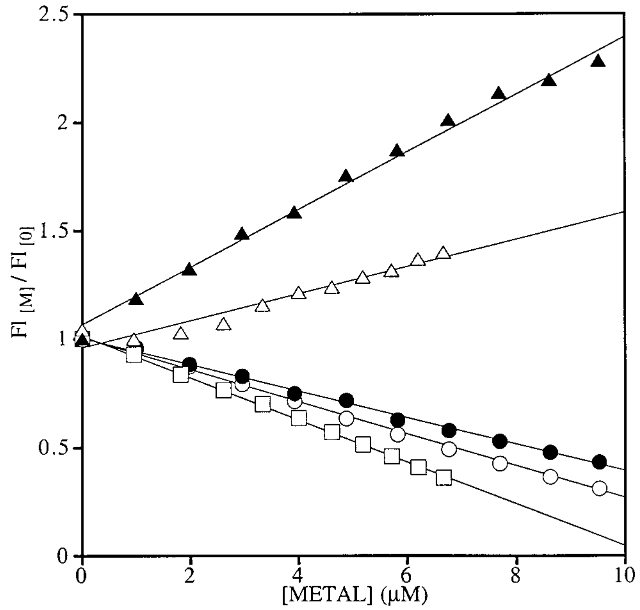

FIG. 4. Fluorescence versus concentration of metal added to Fura-2. The fraction of original fluorescence ($Fl_{[M]}/Fl_0$) is plotted as a function of metal ion concentration in 20 μM Fura-2, 4 M guanidine–HCl, 7.5 mM Mops, pH 7.0 at 25°C. Ni^{2+} overlays with Cu^{2+} and so has been eliminated for clarity. The wavelengths chosen for the assays were λ_{Ex} = 340 (Mn^{2+}, ○; Co^{2+}, ●; Cu^{2+}, □; Ni^{2+}, not shown), 373 (Cd^{2+}, ▲), or 380 nm (Zn^{2+}, △) and λ_{Em} = 510 (Mn^{2+}, Co^{2+}, Cu^{2+}, Zn^{2+}), 500 (Ni^{2+}), or 465 nm (Cd^{2+}). For all of the metals, the fluorescence is linearly dependent on the concentration of added metal with the following parameters: Mn^{2+} ($m = -0.074$, $R = 0.998$), Co^{2+} ($m = -0.063$, $R = 0.998$), Cu^{2+} ($m = -0.097$, $R = 0.999$), Zn^{2+} ($m = 0.063$, $R = 0.967$), Cd^{2+} ($m = 0.133$, $R = 0.996$).

With the weaker binding metals, such as Ni^{2+} and Cd^{2+}, DPA was not essential for catalysis of metal ion equilibration; however, trace amounts of DPA were included in all experiments to maintain a consistent metal ion buffer. The free metal ion concentration was maintained in these experiments using metal ion buffers. Furthermore, the total concentration of metal ions in the dialysis buffers was >40-fold higher than the total protein concentration to prevent any significant alteration in free metal ion concentration due to sequestration by the protein. After equilibration, the protein-bound metal ions were separated from the excess metal ions in solution using a gel filtration column. One potential limitation of this method is that protein-bound metal ions may dissociate from the protein during the 2.5-min column transit time; however, in both cases, the $[E \cdot M]/[E]_{tot}$ ratio at saturating metal extrapolates to 1 ± 0.1. This indicates that metal dissociation during the gel filtration separation was negligible. Finally, 4 M guanidine–HCl was included in the metal ion assays to denature CA II (25) and thereby decrease the metal ion affinity of this protein (33). This allows the lower affinity metallochromic indicators, PAR or Fura-2, to coordinate the released metal ions. The guanidine–HCl may be replaced with any other method (e.g., urea, organic solvents, or heat treatment) that causes a release of metal ions from the protein, so long as the method does not sequester the metal ions or introduce excessive adventitious metal ions.

Using this method, we measured the dissociation constants of CA II for Ni^{2+} and Cd^{2+} (Fig. 5, 16 ± 5 and 2.3 ± 0.3 nM, respectively). The Zn^{2+} affinity of CA II has previously been measured as 0.8 pM (33) under the same conditions. These data demonstrate that CA II binds Zn^{2+} more tightly than Ni^{2+} or Cd^{2+} by more than 17,000- and 2000-fold, respectively. These specificities are much greater than those measured for human CA I isozyme at pH 5.5 (10- and 20-fold, respectively) (10).

DISCUSSION

Transition metal assays. The assays developed here are highly sensitive and capable of measuring concentrations in the low micromolar range. Since the change in absorbance is as high as 0.05 per micromolar metal for a 1-cm path length, we can reliably measure concentrations as low as 0.2 μM. Moreover, small assay

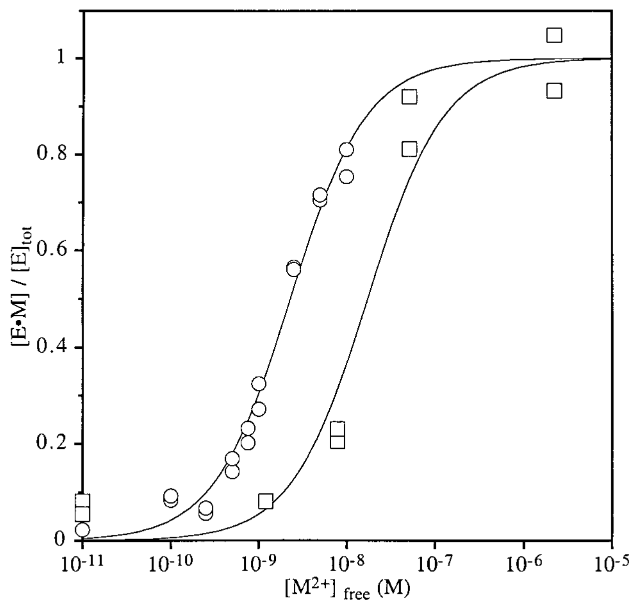

FIG. 5. Measurement of the affinity of human CA II for Cd^{2+} and Ni^{2+}. The metal ion dissociation constant was obtained by dialyzing 0.5 ml of 80 μM CA II for 16–20 h at 30°C against 500 ml of cadmium nitrate (○) or nickel nitrate (□) (each 10 μM to 1 mM total concentration). Buffers (0–15 mM citrate, 0–15 mM Mops, 4–6 μM dipicolinate) were added to maintain both the free metal ion concentration (calculated from the metal ion stability constants (24)) and the pH. Enzyme-bound metal ($E \cdot M$) was separated from free metal by chromatography on gel filtration columns and quantitated using either the PAR colorimetric (Ni^{2+}) or Fura-2 fluorimetric (Cd^{2+}) assay, as described. The data have been adjusted in the following manner: the background has been subtracted from each data set and the endpoints, which ranged from 0.98 to 1.04, have been normalized to 1. The data were fit to Eq. [1].

volumes (0.1–0.5 ml) can be used to determine small total amounts of analyte (20–100 picomoles) without compromising accuracy.

In addition to higher sensitivity, these assays offer a number of advantages which increase their utility. First, this is a rapid in-solution assay with no extensive sample preparation necessary and requiring low sample volumes. Second, each of these assays may be used for a range of biologically interesting metal ions and, furthermore, the ability to use a single assay for a wide range of different metal ions increases the comparability of the resulting measurements. Finally, the equipment required is simple; a spectrophotometer or luminescence spectrophotometer is nearly standard equipment in biochemistry laboratories. Therefore, the development of this pair of closely related simple, rapid, in-solution assays should be of great benefit to the study of the metal ion affinities of metalloproteins and metal ion concentrations in biological samples.

These assays are comparable in detection limit to inductively coupled plasma emission spectroscopy (ICPES) for the first-row transition metal ions. Although both inductively coupled plasma–high-resolution mass spectrometry (ICP–HRMS) and graphite furnace atomic absorption spectroscopy (AA) can detect these ions in the range of 10^{-10}–10^{-11} M or better (i.e., ppb or better), the assays developed here have the advantage of requiring neither the extensive sample preparation necessary for AA nor the extensive sample preparation and large volume required by ICP–HRMS (minimum sample size ~100 ml). Furthermore, both AA and ICP–HRMS require expensive equipment and longer assay times.

A limitation of these colorimetric assays is the possibility of competition for the analyte by contaminating chelators. Both PAR and Fura-2 (22) are weaker chelators than many native metalloenzymes (10, 34, 35). For this reason, denaturing conditions are used in the assays, rendering the protein a weaker chelator. Unfortunately, there is no method to similarly decrease the affinity of possible competing small-molecule chelators, such as EDTA, without affecting the affinity of the indicators. Therefore, contamination of a sample by small-molecule chelators may interfere with the assay. Also, contamination of a sample by competing metal ions may interfere. Both PAR and Fura-2 respond in a similar fashion to a range of different metal ions. Although there are slight differences in the spectra of the various PAR–metal complexes and very slight differences in the spectra of the various Fura-2–metal complexes, these differences are not sufficient to easily discriminate between ions.

Finally, since both the PAR and Fura-2 assays rely on optical detection methods, high sample turbidity may also interfere with detection. Fortunately, the PAR assay relies on detecting changes in the 500-nm range, a wavelength region in which few biomolecules absorb, reducing the possibility of interference from absorption. Furthermore, for the fluorometric assay, ratiometric methods may be used to control for the affects of turbidity or scattering (22).

Metal specificity of CA II. In demonstrating the effectiveness of these assays, we determined the affinity of CA II, a prototypical zinc enzyme, for Ni^{2+} and Cd^{2+}. As predicted, CA II is highly specific for Zn^{2+} and is much more selective for Zn^{2+} over Ni^{2+} or Cd^{2+} (by >17,000-fold and >2000-fold, respectively) than most small-molecule chelators (e.g., EDTA binds Ni^{2+} with 80-fold higher affinity and Cd^{2+} with equal affinity to Zn^{2+}) (24). Although there are very little data on the specificity of other metalloproteins, CA II is much more selective for Zn^{2+} over Ni^{2+} or Cd^{2+} as compared to carbonic anhydrase I at low pH (Zn^{2+} binding is 10-fold better than Ni^{2+} and 20-fold better than Cd^{2+}), carboxypeptidase A (200-fold better than Ni^{2+} but 2-fold weaker than Cd^{2+}) (10), and *Zymomonas mobilis* iron-activated alcohol dehydrogenase (Zn^{2+} affinity is only 4-fold higher than Ni^{2+} affinity) (36).

In vivo assays. Fura-2 has widely been used as an indicator to measure calcium concentrations in complex media and *in vivo.* However, both our data and data from other laboratories (21, 22, 29–31) demonstrate that Fura-2 can coordinate a variety of divalent cations (Ca^{2+}, Mn^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Zn^{2+}, Cd^{2+}, and Pb^{2+}) with reasonable affinity and this complexation affects the observed fluorescence of Fura-2. As previously discussed, the major effect of metal coordination by Fura-2 is to enhance or quench the fluorescence intensity without affecting the peak shape or wavelength greatly. Therefore, although the direction of the fluorescence change or the slight spectral shifts may aid metal identification in pure solutions, the identity of metal–Fura-2 complexes cannot be determined from the spectral properties of complex or mixed samples. Some potential interferents, such as Mg^{2+}, have millimolar intracellular concentrations (37) and are often accounted for in Fura-2 assays for Ca^{2+}. However, despite the low concentrations *in vivo,* the high Fura-2 stability constants with transition metals may lead to interference with Ca^{2+} determination using this indicator. Zinc, for example, is likely to interfere with the measurement of Ca^{2+} concentration since the affinity of Fura-2 for Zn^{2+} is nearly 100-fold higher than its affinity for Ca^{2+} and the spectral responses of the Fura-2 complexes of Zn^{2+} and Ca^{2+} are virtually the same (22). Although the concentration of zinc is typically low (e.g., 1–32 pM for erythrocytes (38)), the concentration in cells may vary widely (e.g., from 20 to 50 μM for splenocytes and thymocytes at rest (39) and up to 0.1 mM concentration in the extracellular space between neurons (40)), reaching concentrations certain to distort Ca^{2+} measurements. Until truly selective indicators are developed, caution must be used in the interpretation of data from complex media. This need for a selective indicator has led to the design of both small-molecule chelators which adopt specific geometries (41) and biosensors.

Biosensors. By exploiting the remarkable specificity of biomolecular recognition, biosensors may provide highly selective indicators. CA II not only has very high specificity for Zn^{2+} but also has higher affinity for Zn^{2+} than many metallofluorescent indicators suggested for use as zinc indicators, including Newport Green (1.0 μM), BTC-5N (coumarin benzothiazole-5N, 100 nM), Mag-fura-2 (20 nM), Fura-2 (3 nM), and TSQ (*N*-(6-methoxy-8-quinolyl)-*p*-toluenesulfonamide, 0.5 nM) (42–44). Therefore, fluorimetric assays using CA II as the receptor molecule have been developed (45, 46). Furthermore, by combining a number of variants with different affinities, a composite biosensor can be prepared that is capable of measuring zinc concentration with high accuracy and specificity over a concentration range spanning several orders of magnitude (47).

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